Quadratic Equation: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Roots and nature of roots

Formulas

$$ax^2 + bx + c = 0 \ (a \neq 0) \ \Rightarrow \ x = \frac{-b \pm \sqrt{D}}{2a}, \qquad D = b^2 - 4ac$$

KEY POINTS

- $D > 0$: two real, distinct roots
- $D = 0$: two real, equal roots
- $D < 0$: imaginary roots (a conjugate pair $p \pm iq$ when a, b, c are real)
- Real roots mean $D \geq 0$
- a, b, c rational and D a perfect square: the roots are rational
- a, b, c rational and one root is $p + \sqrt{q}$: the other root is $p - \sqrt{q}$

Watch out

'Distinct roots' needs $D > 0$. 'Real roots' allows $D \geq 0$.
Check $a \neq 0$ first.

2. Sum, product and difference of roots

Formulas

$$\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}, \qquad |\alpha - \beta| = \frac{\sqrt{D}}{|a|}$$

Cubic $ax^3 + bx^2 + cx + d = 0$: $\sum \alpha = -\frac{b}{a}, \quad \sum \alpha\beta = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$

Watch out

The sum is $-\frac{b}{a}$, not $\frac{b}{a}$.

3. Symmetric expressions

KEY POINTS

- An expression is symmetric if swapping α and β leaves it unchanged.
- Symmetric: $\alpha^2 + \beta^2$, $\frac{1}{\alpha} + \frac{1}{\beta}$. Not symmetric: $\alpha^2 - \beta^2$, $\alpha^2\beta$.

STEPS

1. Find $s = \alpha + \beta$ and $p = \alpha\beta$.
2. Rewrite the expression using s and p (formulas below).
3. Substitute the values.

Formulas

$$\alpha^2 + \beta^2 = s^2 - 2p \qquad \alpha^3 + \beta^3 = s^3 - 3ps \qquad \alpha^4 + \beta^4 = (s^2 - 2p)^2 - 2p^2$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{s}{p} \qquad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{s^2 - 2p}{p} \qquad (\alpha - \beta)^2 = s^2 - 4p$$

With $t = x + \frac{1}{x}$: $x^2 + \frac{1}{x^2} = t^2 - 2, \quad x^3 + \frac{1}{x^3} = t^3 - 3t$
Complex conjugate roots ($D < 0$): $\alpha = re^{i\theta}$, so $\alpha^n + \beta^n = 2r^n \cos n\theta$

4. A relation between the roots is given

KEY POINTS

- Typical relations: $\alpha = 2\beta$, $\alpha - \beta = k$, $\alpha = \beta^2$, a given ratio.
- Integer or natural roots: list the factor pairs of the product and keep the pair with the required sum.

STEPS

1. Write the relation.
2. Write $\alpha+\beta=-\frac{b}{a}$ and $\alpha\beta=\frac{c}{a}$.
3. Eliminate α and β to find the unknown.

Watch out

If real roots are required, check that your answer gives $D \geq 0$.

5. Forming an equation

Formulas

$$x^2-(\alpha+\beta)x+\alpha\beta=0 \qquad x^3-S_1x^2+S_2x-S_3=0 \qquad ax^2+bx+c=a(x-\alpha)(x-\beta)$$

KEY POINTS

- Real coefficients and a root $p+iq$: the other root is $p-iq$.
- Rational coefficients and a root $p+\sqrt{q}$: the other root is $p-\sqrt{q}$.
- To get $(k-\alpha)(k-\beta)$, put $x=k$ in $\frac{f(x)}{a}$.
- If α is a root, replace α^2 by a linear expression in α again and again to simplify.
- Perfect square: $a>0$ and $D=0$.

6. New equation whose roots are $f(\alpha),\ f(\beta)$

STEPS

1. Put $y=f(x)$.
2. Solve for x in terms of y.
3. Substitute in the given equation and simplify.
4. Shortcut: if the sum S and product P of the new roots are easy, write $x^2-Sx+P=0$.

KEY POINTS

- Roots $k\alpha$: replace x by $\frac{y}{k}$.
- Roots $\alpha+k$: replace x by $y-k$.
- Roots $\frac{1}{\alpha}$: replace x by $\frac{1}{y}$ (the coefficients reverse).
- Roots α^2: use S and P, or remove the square root by squaring.

7. Newton's formula (powers of roots)

Formulas

$$S_n=p\alpha^n+q\beta^n \ \Rightarrow\ aS_{n+2}+bS_{n+1}+cS_n=0$$

STEPS

1. Write the recurrence from the equation.
2. Use it to reduce high powers to low powers.
3. The expression in the question usually collapses to a number.

Watch out

Take care of the sign of b when you write the recurrence.

8. Special cases and identity

KEY POINTS

- Exactly one root 0: $c=0,\ b\neq 0$.
- Both roots 0: $b=0,\ c=0$.
- Exactly one root infinite: $a=0,\ b\neq 0$.
- Both roots infinite: $a=0,\ b=0,\ c\neq 0$.
- Identity (true for every x): $a=b=c=0$. A quadratic with more than 2 roots is an identity.
- $a+b+c=0 \Rightarrow x=1$ is a root (the other root is $\frac{c}{a}$).
- $a-b+c=0 \Rightarrow x=-1$ is a root.

Watch out

Do not drop the 'not equal to zero' parts ($b \neq 0$, $c \neq 0$).

9. Common roots

Formulas

Both roots common: $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$

Exactly one root common: $(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$

STEPS

1. Let α be the common root.
2. Subtract the equations to remove α^2.
3. Find α in terms of the parameter.
4. Put α in one equation to get the parameter.
5. Check every value you get.

KEY POINTS

- If one equation has imaginary roots and real coefficients, a common root forces both roots to be common (conjugate pair).

10. Quadratic graph, sign and range

Formulas

Vertex $\left(-\dfrac{b}{2a}, -\dfrac{D}{4a}\right)$; axis $x = -\dfrac{b}{2a}$; y-intercept c

Range: $a > 0$: $\left[-\dfrac{D}{4a}, \infty\right)$ $a < 0$: $\left(-\infty, -\dfrac{D}{4a}\right]$

Always positive: $a > 0,\ D < 0$. Always negative: $a < 0,\ D < 0$. (Use $D \leq 0$ for ≥ 0 or ≤ 0.)

KEY POINTS

- Reading a graph: opens up means $a > 0$; the y-intercept gives c.
- The graph cuts, touches or misses the x-axis: $D > 0,\ D = 0,\ D < 0$.
- If $f(p) = f(q)$ with $p \neq q$, the axis is $x = \frac{p+q}{2}$.
- Range by discriminant: write $ax^2 + bx + (c - y) = 0$ and use $D \geq 0$.

STEPS

1. Restricted $x \in [p, q]$: find the vertex x-value $-\frac{b}{2a}$.
2. If it lies inside $[p, q]$, one extreme value is at the vertex.
3. Compare with the values at p and q.

11. Range of rational functions

Formulas

$y = \dfrac{ax + b}{cx + d}$: range $\mathbb{R} - \left\{\dfrac{a}{c}\right\}$

STEPS

1. For $\frac{Q}{Q}$, $\frac{Q}{L}$, $\frac{L}{Q}$: put $y = f(x)$ and cross-multiply to a quadratic in x.
2. Case 1: leading coefficient $\neq 0$: use $D \geq 0$ and solve for y.
3. Case 2: leading coefficient $= 0$: check whether that y is attained.
4. Join both cases.

KEY POINTS

- Common factor: cancel it, find the range, then remove the y-value at the cancelled x.
- $\frac{Q}{L}$: the leading coefficient is a constant, so there is no Case 2.
- x restricted: solve $x = g(y)$ and put the condition on x.

Watch out

Forgetting Case 2 is the most common mistake.

12. Remainder theorem and Descartes' rule

KEY POINTS

- $f(x) \div (x-a)$: remainder $f(a)$. $(x-a)$ is a factor when $f(a)=0$.
- Divisor $(x-a)(x-b)$: remainder $ux+v$; find u, v from $f(a)$ and $f(b)$.
- Descartes: the number of positive roots is at most the number of sign changes in $f(x)$, and differs from it by an even number.
- For negative roots apply the rule to $f(-x)$.

13. Location of roots

KEY POINTS

- Three tools: D, the vertex $-\frac{b}{2a}$ against the given numbers, and the sign of $af(p)$.

Formulas

Both roots $> d$: $D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} > d$
Both roots $< d$: $D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} < d$
Both roots in (d, e): $D \geq 0,\ af(d) > 0,\ af(e) > 0,\ d < -\frac{b}{2a} < e$
d between the roots: $af(d) < 0$ (no D needed)
One root $< d$ and the other $> e$: $af(d) < 0,\ af(e) < 0$
Exactly one root in (d, e): $f(d)f(e) < 0$
One root +ve, one root −ve: $af(0) < 0$
Both roots +ve: $D \geq 0,\ -\frac{b}{2a} > 0,\ af(0) > 0$
Both roots −ve: $D \geq 0,\ -\frac{b}{2a} < 0,\ af(0) > 0$

STEPS

1. Exactly one root in (d, e): write $f(d)f(e) < 0$.
2. Test $f(d) = 0$ separately: find the other root and see if it is inside.
3. Test $f(e) = 0$ the same way.

KEY POINTS

- $f(p)f(q) < 0$ means a root in (p, q) (Intermediate Value Theorem).
- $x^4 + px^2 + q = 0$: put $t = x^2$. Each $t > 0$ gives 2 values of x, $t = 0$ gives 1, $t < 0$ gives none.
- At least one positive root $(a > 0)$: $f(0) < 0$, or $(D \geq 0,\ -\frac{b}{2a} > 0,\ f(0) \geq 0)$.

14. Higher degree equations

Formulas

$a_0x^n + a_1x^{n-1} + \cdots + a_n = 0$: $S_k = (-1)^k \dfrac{a_k}{a_0}$ (sum of products of roots taken k at a time)

$$\sum \alpha_i^2 = S_1^2 - 2S_2$$

$$\alpha + \beta + \gamma = 0 \ \Rightarrow\ \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

KEY POINTS

- Factor form $f(x) = a_0(x - x_1)\cdots(x - x_n)$: put a number for x to get products such as $\prod(1 - x_i)$.

15. Inequalities and modulus

STEPS

1. Wavy curve: make the right side 0 and the leading coefficients positive.
2. Factorise into linear factors.
3. Mark the zeroes (exclude zeroes of the denominator).
4. Start with + on the right.
5. Change sign only across factors with an odd power.

KEY POINTS

- Double inequality: split into two, solve each, take the intersection.
- $|q| < k \iff -k < q < k$. $|q| > k \iff q < -k$ or $q > k$.
- $|q(x)| = \ldots$: split at the zeroes of q and keep only roots inside their interval.
- One solution set inside another: $[r_1, r_2] \subseteq [s_1, s_2] \iff r_1 \geq s_1,\ r_2 \leq s_2$. Check each case of the parameter and the empty set.
- $f(x) > 0$ for all $x > 0$ ($a > 0$): $D < 0$, or ($D \geq 0,\ -\frac{b}{2a} \leq 0,\ f(0) \geq 0$).

16. Second degree in two variables

Formulas

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

Two linear factors: $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ (and $h^2 \geq ab$ for real factors)
$ax^2 + 2hxy + by^2 > 0$ for all $(x, y) \neq (0, 0) \iff a > 0$ and $h^2 < ab$

STEPS

1. Range of y: treat the equation as a quadratic in x and apply $D_x \geq 0$.
2. Range of x: treat it as a quadratic in y and apply $D_y \geq 0$.

KEY POINTS

- Factorising: make D a perfect square in the other variable, or factorise the x^2, xy, y^2 part and add constants c_1, c_2.

17. Substitutions and standard tricks

KEY POINTS

- $t = x + \frac{1}{x} \Rightarrow |t| \geq 2$. $t = x^2 \Rightarrow t \geq 0$. $t = e^x$ or a^x $(a > 0) \Rightarrow t > 0$.
- Reciprocal equation $ax^4 + bx^3 + cx^2 \pm bx + a = 0$: divide by x^2; $t = x + \frac{1}{x}$ (for $+bx$) or $t = x - \frac{1}{x}$ (for $-bx$).
- Homogeneous in two expressions u, v: divide by v^2, put $t = \frac{u}{v}$, then solve $u = tv$.
- Product of four linear factors $= k$: pair them so the constants have equal sum; the common quadratic is t.
- $p(p(x)) = 0 \iff p(x) = r$, where r is a root of p.
- Continued radical or fraction: $x = \sqrt{a + x}$ or $x = a + \frac{1}{x}$; reject the root that breaks the sign condition.
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$; it is 0 when $a + b + c = 0$ or $a = b = c$.

Watch out

Squaring can add extra roots: substitute back and check the sign of every square root.
After a substitution, restrict t to its allowed values before going back to x.

18. Common mistakes checklist

STEPS

1. Forgetting $a \neq 0$, or using $D \geq 0$ when distinct roots ($D > 0$) are needed.
2. Wrong sign in Vieta's formulas: the sum is $-\frac{b}{a}$.
3. Skipping Case 2 (leading coefficient zero) in range questions.
4. Not checking the roots after squaring.
5. In location of roots: forgetting the vertex condition when both roots are on one side, or not testing the end points.
6. In the wavy curve: changing sign across an even power, or including a zero of the denominator.
7. After $t = x + \frac{1}{x}$, e^x or x^2: not restricting t.
8. Not checking that a parameter value really gives real roots.
9. In common-root questions: not verifying every value of the parameter.